Literature Summary Report

Bangladesh-Contextualized Cloud Forensics: Banking Sector, Legal Landscape, and Jurisdictional Constraints

Prepared to support: "Toward a Cloud Forensic Readiness Framework for Bangladesh's Banking Sector"

MD. Ahsan Kabir (ID 0122520016) · MSCSE, United International University · CSE 6093 Research Methodologies and Technical Writing

Legend: **[Peer-reviewed]** **[Primary legislation / regulation]** **[Government report / grey literature]**

Each entry below was individually verified (via Crossref, publisher pages, or by reading the source directly) unless otherwise flagged. Sources flagged as unverifiable in the research pass are excluded from this report.

1. Bangladesh's Legal & Institutional Digital-Evidence Landscape

Evidence (Amendment) Act, 2022 (Act No. XX of 2022)

Government of Bangladesh, Ministry of Law · 2022 **[Primary legislation]**

Made digital records admissible in Bangladeshi courts for the first time, via a Section 65B certificate requirement modeled on the Indian Evidence Act amendment. Defines "digital record" broadly (computer memory, CCTV footage, drone data, cell phone records, digital footprints) by cross-reference to the Digital Security Act 2018. Critically, the certificate mechanism assumes a clear, single custodian who can attest to the record's origin and integrity — a condition that breaks down when evidence is a log entry on a foreign hyperscale cloud server with no Bangladeshi legal presence.

http://bdlaws.minlaw.gov.bd/act-details-1416.html

Cyber Security Act, 2023

Government of Bangladesh · 2023 (repeals the Digital Security Act 2018) **[Primary legislation]**

Sections 9-11 establish emergency response mechanisms, mandate digital forensic labs, and set quality-control standards for forensic work; separately imposes duties on regulated entities to preserve forensic evidence and report breaches to the national CIRT. Retains much of the DSA 2018's regulatory structure while reducing penalty severity. Provides the current statutory basis for forensic lab operation in Bangladesh, but — like the Evidence Act — does not address cross-border cloud evidence specifically.

(Act text via Bangladesh Gazette; summarized in ICNL handbook below)

Research Monograph on Cybercrime and Digital Evidence in Bangladesh: Problem and Prospect

Md Mostafa Kamal Uzzal · SSRN working paper, 12 Jan 2025 · DOI: 10.2139/ssrn.5876822 **[Government report / grey literature]**

The single most directly on-point Bangladeshi source located. Surveys the DSA 2018, ICT Act 2006, and amended Evidence Act 2022 alongside institutional responses from CID, the Counter Terrorism and Transnational Crime unit (CTTC), and the Digital Security Agency. Identifies legal ambiguity around digital evidence handling, procedural gaps, limited forensic capacity, and weak inter-agency coordination — the exact institutional weaknesses a cloud forensic readiness framework would need to work around. Not peer-reviewed; cite as a working paper.

https://ssrn.com/abstract=5876822

A Study on Forensic Services in Criminal Investigation in Bangladesh: Impacts, Challenges and Capacity Building Issues

Barkat, A. et al. (9 co-authors) · Human Development Research Centre (HDRC), Dhaka, Feb 2020 · commissioned by Bangladesh Police CID **[Government report / grey literature]**

An 89-page commissioned report, two of whose co-authors are serving officers in CID's Digital Forensics & Cyber Police Centre. Chapter 8 documents lab operations and capacity; Figure 8.1 charts the growing backlog of

digital forensic cases; Section 10.3.11 is a dedicated discussion of Digital Forensic Lab challenges, tying the backlog directly to human-resource shortages in the Dhaka and Chattogram labs and resulting judicial delays. The strongest available evidence that Bangladesh's domestic forensic capacity is a binding constraint, independent of any cloud-specific issue.

https://www.hdrc-bd.com/wp-content/uploads/2020/09/33.-Final-Report_Forensic-Services-in-Criminal-Investigation-Department-CID_2020.pdf

Digital Forensic Lab Guidelines, Version 02

ICT Division, Government of Bangladesh · 17 May 2021 **[Primary legislation]**

The national standard for digital forensic lab practice in Bangladesh. Establishes procedural expectations for evidence handling within domestic labs. Notably scoped to physical/on-premises evidence; does not address acquisition from cloud-hosted or foreign-jurisdiction sources, reinforcing the gap this proposal targets.

ictd.portal.gov.bd (retrieve via browser — host currently serves a mismatched TLS certificate)

BGD e-GOV CIRT (Bangladesh Government Computer Incident Response Team)

Bangladesh Computer Council · ongoing **[Government report / grey literature]**

Bangladesh's designated national CERT, established in direct response to the 2016 Bangladesh Bank cyber heist — a clean institutional link between this proposal's case study and its regulatory landscape. Offers digital forensics, vulnerability assessment/penetration testing, and IT audit services, and recently launched a National ICT & Cybersecurity Rating System for maturity assessment across sectors including finance.

https://www.cirt.gov.bd/who-we-are

Personal Data Protection Ordinance (PDPO) 2025

Government of Bangladesh · enacted November 2025 **[Primary legislation]**

A major, very recent development. Reported (via secondary sources; the gazette text should be obtained directly) to impose cloud data-localization under an amendment to Section 29(7)(b): restricted personal data and critical-infrastructure-related data must maintain at least one synchronized real-time copy inside Bangladesh, with "confidential" or "restricted" classifications required to be stored in-country entirely. This directly intersects with Bangladesh Bank's cloud computing guideline (below) and materially changes the legal environment this proposal analyzes.

https://itif.org/publications/2025/05/16/bangladesh-cross-border-data-transfer-regulation/ · https://www.thedailystar.net/tech-startup/news/bangladeshs-personal-data-protection-ordinance-2025-key-takeaways-4015401

Technological Advancement and the Rise of Cybercrime in Bangladesh: Trends, Challenges, and Policy Responses

Md. Mahbubur Rahman & Md. Shahidulla Kaiser · in The Palgrave Handbook of Global Social Problems, Springer Nature, 2025 · DOI: 10.1007/978-3-030-68127-2_722-1 **[Peer-reviewed]**

A peer-reviewed book chapter situating Bangladesh's cybercrime trends within broader social-problems scholarship. Provides policy-level framing useful for the proposal's introduction and for connecting technical forensic gaps to national policy discourse.

DOI: 10.1007/978-3-030-68127-2_722-1

2. The 2016 Bangladesh Bank Heist — Anchor Case Study

Institutional Fragility and Cross-Border Cyber Risk in Financial Systems: Insights from the 2016 Bangladesh Bank Heist

M. Sirajul Islam · Computers (MDPI), 15(7), art. 432, July 2026 · DOI: 10.3390/computers15070432 **[Peer-reviewed]**

The single best anchor paper available: recent, open-access, and explicitly analyzes the heist as a developing-country institutional-fragility problem rather than a purely technical intrusion. Uses a scoping review plus exploratory case study across macro (institutional/geopolitical), meso (ecosystem interdependency), and micro (organizational) levels, attributing the incident to fragmented governance, digital dependencies, limited

operational resilience, and human-factor vulnerabilities. Directly supports framing the heist as a cross-border cloud/financial-infrastructure governance failure.
https://www.mdpi.com/2073-431X/15/7/432

The Spillover Effect of the Bangladesh Bank Cyber Heist on Banks' Cyber Risk Disclosures in Bangladesh

Mohammed Mehadi Masud Mazumder & Abdus Sobhan · Journal of Operational Risk, 15(4), Dec 2020, pp. 53-76 · DOI: 10.21314/JOP.2020.249 **[Peer-reviewed]**

An institutional-theory study of annual reports from 38 Bangladeshi commercial banks (2014-2018), finding that cyber-risk disclosures rose significantly sector-wide after the heist, though politically connected and Shariah-compliant banks were less likely to use disclosure as a legitimacy strategy. Useful evidence that the heist produced a measurable sector-wide institutional aftershock, supporting the claim that banking-sector cloud forensic readiness is a live, recognized concern rather than a hypothetical one.
https://www.risk.net/journal-of-operational-risk/7736786/

The SWIFT Heist: A Cybersecurity Case

Lorraine S. Lee & Richard D. Mautz III · Journal of Forensic Accounting Research, 10(1), 2025, pp. 335-356 **[Peer-reviewed]**

A peer-reviewed teaching case (not an empirical study — cite accordingly) applying COSO internal-control analysis to the Bangladesh Bank incident. Notes that attackers exploited cultural and time-zone differences between Bangladesh, the New York Federal Reserve, and the Philippines to conceal the nearly $1 billion attempt. Useful for illustrating how cross-border operational and time-zone gaps compound forensic and control challenges.
https://publications.aaahq.org/jfar/article/10/1/335/13927/

3. Cloud Adoption, Regulation, and Infrastructure in Bangladesh

Guidelines on Cloud Computing (BRPD Circular No. 05)

Bangladesh Bank, Banking Regulation and Policy Department · 16 March 2023 **[Primary legislation]**

The central regulatory document for this proposal. Applies to banks, NBFIs, mobile financial service providers, and payment/settlement systems. Requires private cloud infrastructure to be located within Bangladesh and owned/operated by the institution itself; mandates data classification before any public-cloud hosting; and requires prior written Bangladesh Bank approval before hosting sensitive customer data in public or cross-border hybrid cloud. Notably, published summaries of the guideline appear silent on regulator or forensic-investigator audit access to cloud-hosted data during an incident — a gap that, if confirmed in the full text, is itself a publishable finding this proposal can surface.
https://www.bb.org.bd/aboutus/regulationguideline/brpd/guideline_cloud_computing.pdf

Guideline on ICT Security, Version 4.0

Bangladesh Bank · 2023 **[Primary legislation]**

Bangladesh Bank's overarching ICT security guideline for banks and NBFIs, of which the cloud computing circular is a specialized extension. Establishes baseline governance, risk assessment, and incident response expectations against which any proposed forensic readiness framework should be positioned as compatible, not competing.
https://www.bb.org.bd/aboutus/regulationguideline/brpd/guideline_v3_ict.pdf

National Data Center, Bangabandhu Hi-Tech City

Bangladesh Data Center Company Ltd. (BDCCL) / Bangladesh Computer Council · commissioned 2019 **[Government report / grey literature]**

Bangladesh's Uptime Institute Tier IV-certified national data center — widely reported as the world's 7th largest Tier IV facility — offering government cloud (G-Cloud), storage, backup, and co-location services. Establishes that Bangladesh has sovereign data-center capacity but not a hyperscale public cloud offering equivalent to AWS,

Azure, or GCP, which matters directly for where financial-sector cloud workloads can legally and practically reside.
https://ndc.bcc.gov.bd/ · https://www.datacenterdynamics.com/en/news/bangladeshs-first-cloud-data-center-starts-operations/

AWS Outposts Rack Now Available in Bangladesh

Amazon Web Services · November 2022 **[Government report / grey literature]**

Confirms Bangladesh has no AWS Region or Local Zone; the only in-country AWS presence is Outposts — customer-premises hardware managed remotely from the nearest offshore AWS Region. AWS's own marketing frames this as a way to meet data-residency requirements precisely because the actual cloud region is elsewhere. Establishes, concretely, that Bangladeshi organizations using major hyperscale cloud providers have their data processed and logged outside Bangladeshi jurisdiction by default.
https://aws.amazon.com/about-aws/whats-new/2022/11/aws-outposts-rack-service-bangladesh/

4. Digital and Cloud Forensics — Bangladesh-Specific and Comparative

National Digital Forensics Framework for Bangladesh

Mohammad Mahfuzul Haque & Syed Akhter Hossain · IEEE 3rd International Conference on Electrical Information and Communication Technology (EICT), 2017 **[Peer-reviewed]**

The closest existing precedent to a national forensic framework proposal for Bangladesh, covering both proactive and reactive (live and post-mortem) forensics. Predates Bangladesh's meaningful cloud adoption and the 2023 Bangladesh Bank cloud guideline, strengthening the case that an update specifically addressing cloud evidence is overdue rather than duplicative.
https://ieeexplore.ieee.org/document/8275133/

Digital Forensics Empowerment of Bangladesh Police: A Comprehensive Analysis of Its Role in Preventing Crimes and Enhancing Investigative Capabilities

Khadija Sharmin & Mohammad Ishtiaque Rahman · Bangladesh Journal of MIS, 11(1), June 2025 · DOI: 10.61606/BJMIS.V11N1A1 **[Peer-reviewed]**

A survey of 52 Bangladesh Police officers examining six dimensions of digital forensics impact: evidence accuracy, collection procedures, geolocation/criminal mapping, conviction rates, crime prevention, and investigative perspectives. Concludes that officers require substantially more digital-forensics training — direct, current evidence of the human-capacity constraint this proposal must design around, not just work with.
https://bjmis.du.ac.bd/index.php/about/article/view/34

An Insight into Cloud Forensic Readiness by Leading Cloud Service Providers: A Survey

Computing (Springer), 2022 · DOI: 10.1007/s00607-022-01077-2 **[Peer-reviewed]**

Compares what major cloud service providers actually provision for forensic purposes across their platforms. Directly useful for grounding what a Bangladeshi investigator can and cannot realistically obtain from AWS, Azure, or GCP when investigating an incident involving a Bangladeshi financial institution's cloud workload.
https://link.springer.com/article/10.1007/s00607-022-01077-2

Factors Influencing the Adoption of Cloud Incident Handling Strategy: A Preliminary Study in Malaysia

arXiv preprint, not peer-reviewed **[Government report / grey literature]**

The closest available developing-country / Southeast Asian analogue to the proposed Bangladesh study. Useful primarily as a methodological template — a preliminary survey-based study of incident-handling adoption factors in a comparable regional economy — rather than as a source of findings to cite directly.
https://arxiv.org/pdf/1505.02908

5. Cross-Border Jurisdictional Constraints

Budapest Convention on Cybercrime — Bangladesh Country Status

Council of Europe, Octopus Cybercrime Community **[Government report / grey literature]**

Confirms Bangladesh is not a party to the Budapest Convention. Consequence: Bangladesh cannot use Article 29 expedited data-preservation requests, Article 32 direct trans-border access provisions, or the Convention's 24/7 cooperation network — all mechanisms that would otherwise ease cross-border cloud evidence requests.

https://www.coe.int/en/web/octopus/-/bangladesh

United Nations Convention against Cybercrime (Hanoi Convention)

United Nations / UNODC · opened for signature 25-26 October 2025 **[Government report / grey literature]**

Bangladesh did not sign at the Hanoi ceremony (72 states did) nor by the most recent count available (76 signatories as of May 2026, including regional neighbors Sri Lanka and the Maldives). A sharp, current, and citable contrast between Bangladesh and its immediate regional peers on international cybercrime cooperation.

https://www.unodc.org/unodc/frontpage/2025/October/seventy-two-nations-sign-first-un-treaty-to-fight-cybercrime--in-milestone-for-digital-cooperation.html

Mutual Legal Assistance in Criminal Matters Act, 2012

Government of Bangladesh, with Implementation Guideline from the Ministry of Home Affairs **[Primary legislation]**

Bangladesh's domestic MLA framework operates on a reciprocity basis rather than through a bilateral treaty network. No evidence was found of a bilateral United States-Bangladesh MLAT, which is significant given that AWS, Microsoft, Google, and Meta are all US-jurisdiction providers most likely to hold Bangladeshi financial-sector cloud data.

http://bdlaws.minlaw.gov.bd/upload/act/2021-11-17-11-42-18-41.-Mutual-Legal-Assitant-in-Criminal-Matters-Act-2012.pdf

UNODC and Justice & Care: Raising Awareness on Mutual Legal Assistance in Criminal Matters

UNODC South Asia · March 2024 **[Government report / grey literature]**

UNODC capacity-building initiative for Bangladeshi law enforcement on MLA procedure. Reports "minimal utilization of foreign evidence in investigations and trials despite the prevalence of cases with international links" — near-direct empirical confirmation, from the UN itself, of this proposal's central premise that Bangladesh's cross-border evidence-acquisition pathway is underused and under-capacitated in practice, not merely on paper.

https://www.unodc.org/southasia//frontpage/2024/March/bangladesh_-unodc-boost-law-enforcement-capacities-on-mutual-legal-assistance-in-criminal-matters.html

Meta and Google Transparency Reports — Bangladesh Government Data Requests, 2024

Meta Transparency Center; reported via Prothom Alo and The Daily Star, 2024-2025 **[Government report / grey literature]**

Meta reported Bangladesh government requests for data on 3,771 accounts in 2024, with compliance rates of 68.4% (H1) and 64% (H2); Google separately received 5,827 removal requests. Quantifies that roughly a third of Bangladeshi requests for cloud-platform-held evidence fail outright, and that successful requests proceed at the discretion of foreign companies rather than through enforceable treaty obligations — concrete, current numbers supporting the proposal's jurisdictional-gap argument.

https://en.prothomalo.com/bangladesh/vzvhi9wkhf

Digital Evidence and Cloud Forensics: Contemporary Legal Challenges and the Power of Disposal

Christos Karagiannis & Kostas Vergidis · Information (MDPI), 12(5), 181, 2021 **[Peer-reviewed]**

Frames cross-border cloud evidence problems around three conceptual pillars — territoriality (loss of a fixed data location), possession (who legally holds the content), and confiscation procedure. Provides the conceptual scaffolding this proposal applies concretely to the Bangladesh case, rather than treating jurisdiction as an undifferentiated "legal problem."

https://www.mdpi.com/2078-2489/12/5/181

A Note on Excluded Sources

Several sources surfaced during research were deliberately excluded from this report and should not be cited. Three apparently fluent citations (a Kabir et al. 2020 JDFSL paper, a Rahman 2017 IJCSDF paper, and a Rahman et al. 2020

International Journal of Cyber Criminology paper) could not be verified against any real publication and are believed to be fabricated. Several RSIS International journal articles (IJRIAS, IJRSI, IJRISS) were topically relevant but are published by a venue widely regarded as low-quality; use only with your supervisor's explicit approval. A ResearchGate item on Bangladeshi bank cloud accounting adoption and a similarly unverifiable cloud-IoT forensic framework paper from Green University of Bangladesh could not be traced to a confirmed peer-reviewed venue and are noted here as a documented literature gap rather than included as evidence.